

21. Develop a C program to implement the Worst Fit algorithm of memory management.

AIM

To write a C program to implement the **Worst Fit memory allocation algorithm**, where each process is allocated to the largest available memory block.

CODE

```
#include <stdio.h>

int main()
{
    int block[10], process[10];
    int nb, np;
    int i, j, worst;

    printf("Enter number of blocks: ");
    scanf("%d", &nb);

    printf("Enter block sizes:\n");
    for(i = 0; i < nb; i++)
        scanf("%d", &block[i]);

    printf("Enter number of processes: ");
    scanf("%d", &np);

    printf("Enter process sizes:\n");
    for(i = 0; i < np; i++)
        scanf("%d", &process[i]);

    for(i = 0; i < np; i++)
    {
        worst = -1;
```

```

for(j = 0; j < nb; j++)
{
    if(block[j] >= process[i])
    {
        if(worst == -1 || block[j] > block[worst])
            worst = j;
    }
}

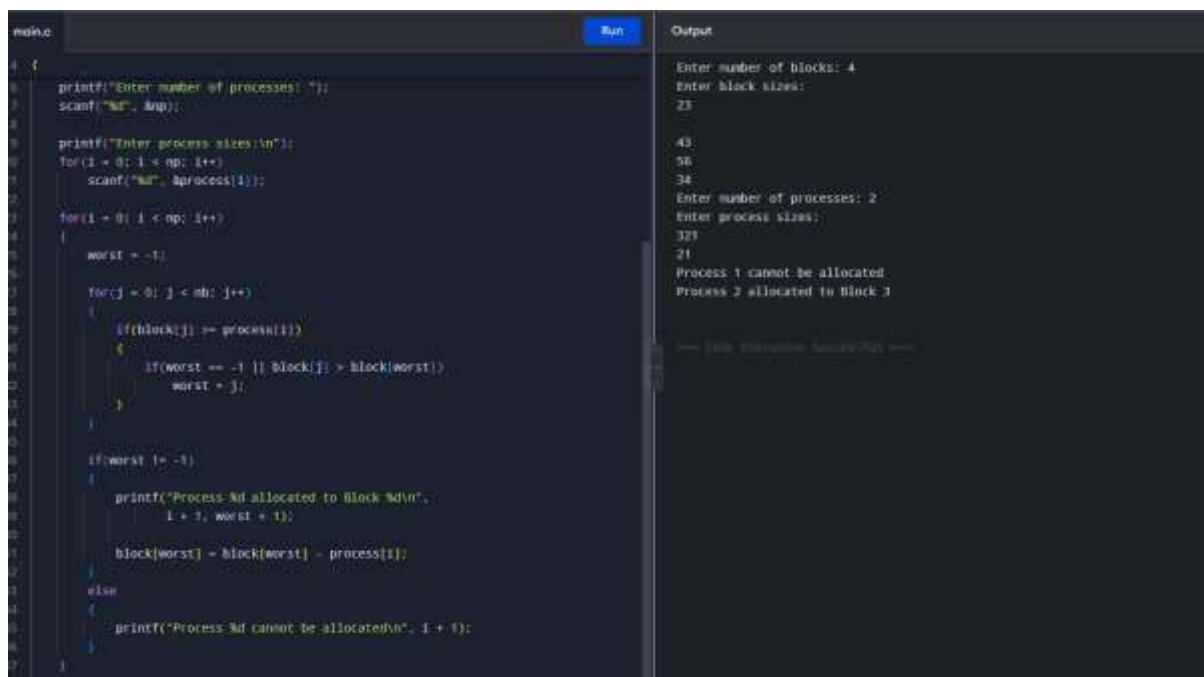
if(worst != -1)
{
    printf("Process %d allocated to Block %d\n",
           i + 1, worst + 1);

    block[worst] = block[worst] - process[i];
}
else
{
    printf("Process %d cannot be allocated\n", i + 1);
}

return 0;
}

```

OUTPUT:



```
main.c Run Output
1 {
2     printf("Enter number of processes: ");
3     scanf("%d", &np);
4
5     printf("Enter process sizes:\n");
6     for(i = 0; i < np; i++)
7         scanf("%d", &process[i]);
8
9     for(i = 0; i < np; i++)
10     {
11         worst = -1;
12
13         for(j = 0; j < nb; j++)
14         {
15             if(block[j] >= process[i])
16             {
17                 if(worst == -1 || block[j] > block[worst])
18                     worst = j;
19             }
20         }
21
22         if(worst != -1)
23         {
24             printf("Process %d allocated to Block %d\n",
25                    i + 1, worst + 1);
26
27             block[worst] = block[worst] - process[i];
28         }
29         else
30         {
31             printf("Process %d cannot be allocated\n", i + 1);
32         }
33     }
34 }
```

Enter number of blocks: 4
Enter block sizes:
23
43
56
34
Enter number of processes: 2
Enter process sizes:
329
21
Process 1 cannot be allocated.
Process 2 allocated to Block 1

RESULT

Thus, the **Worst Fit memory allocation algorithm** was successfully implemented and executed.